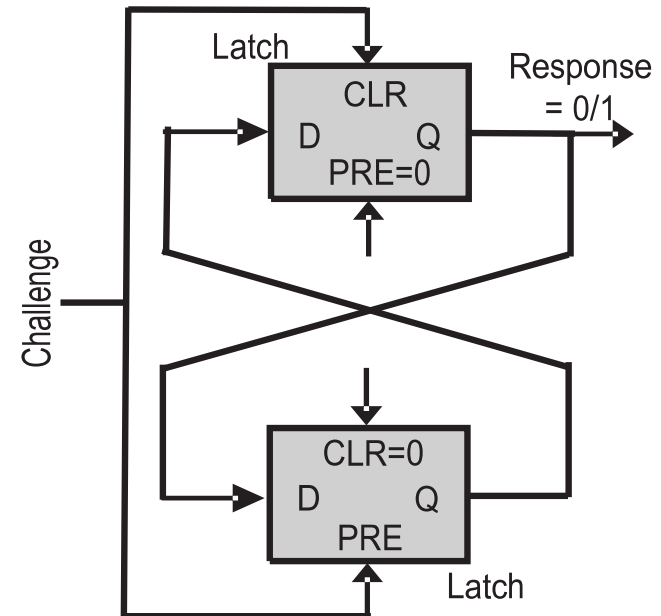
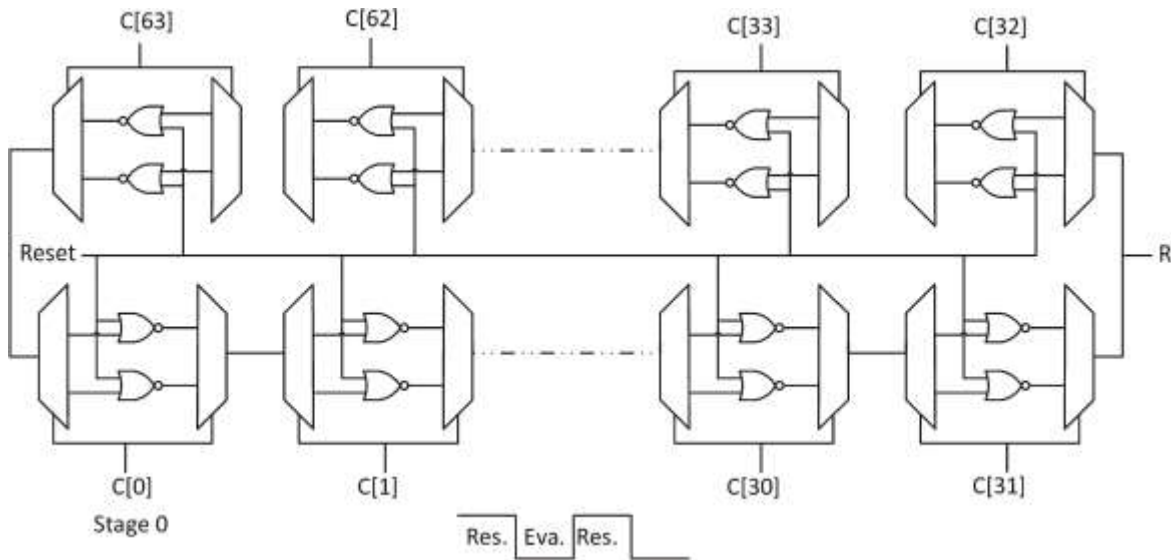
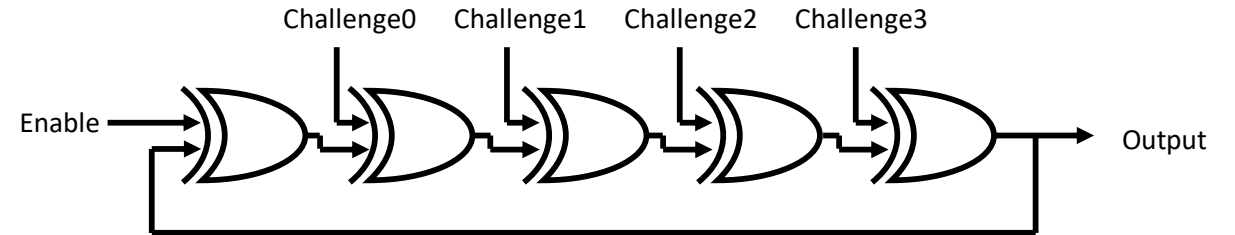
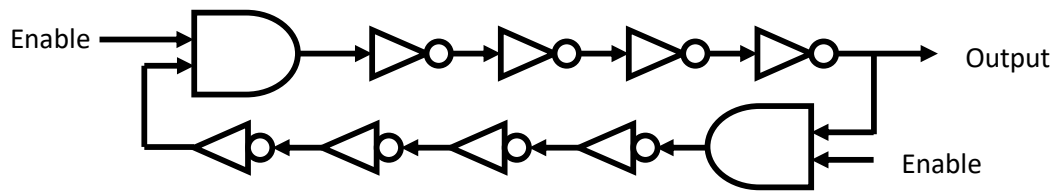
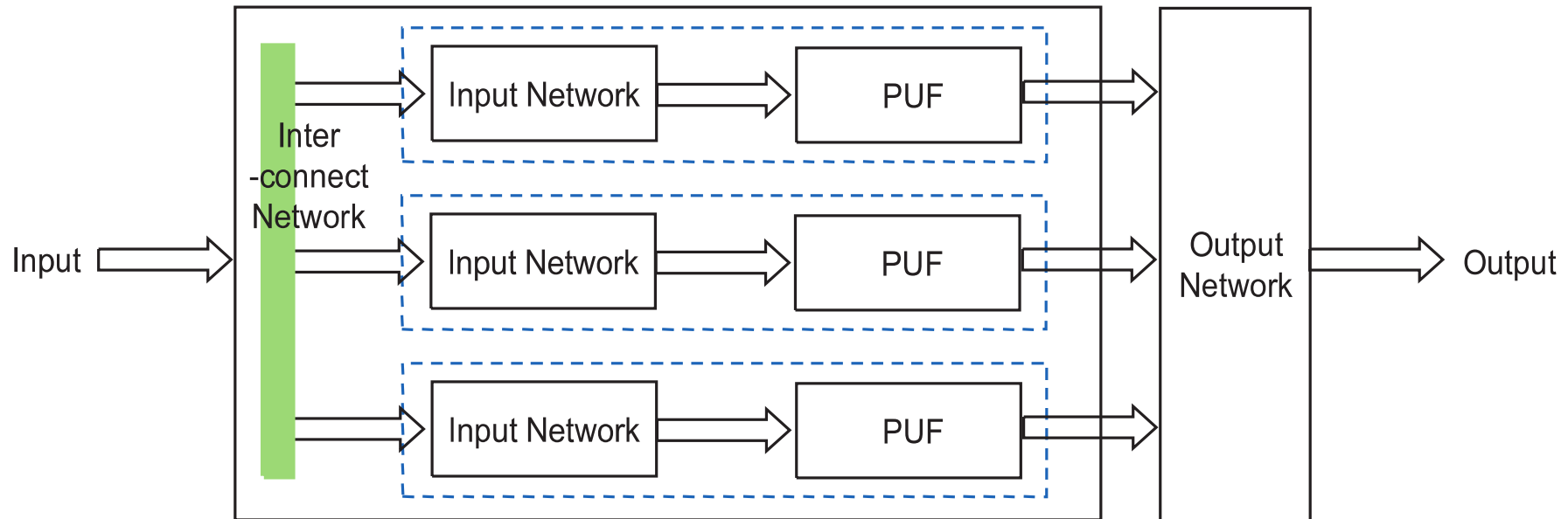


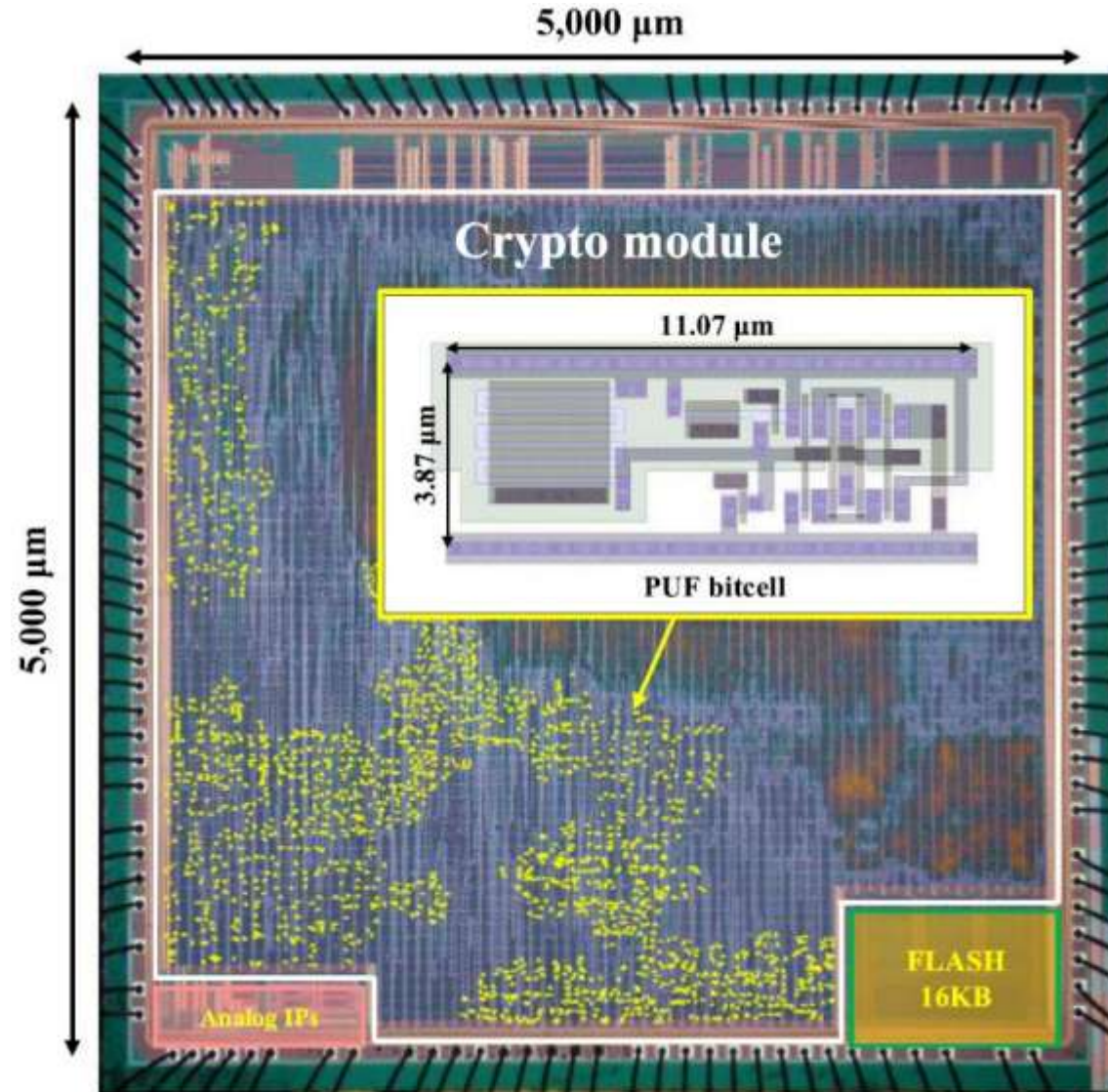
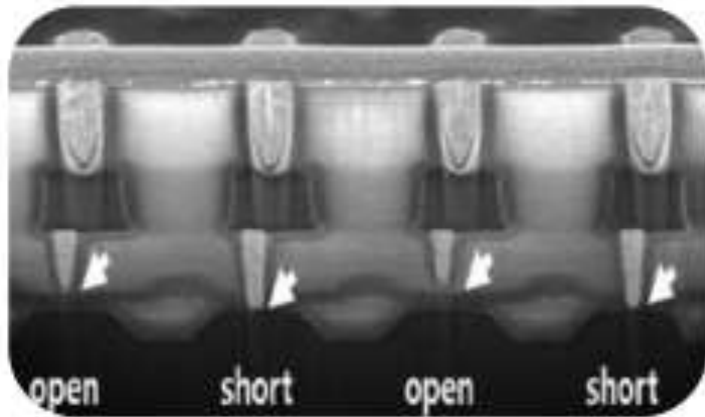
PUF Derivatives



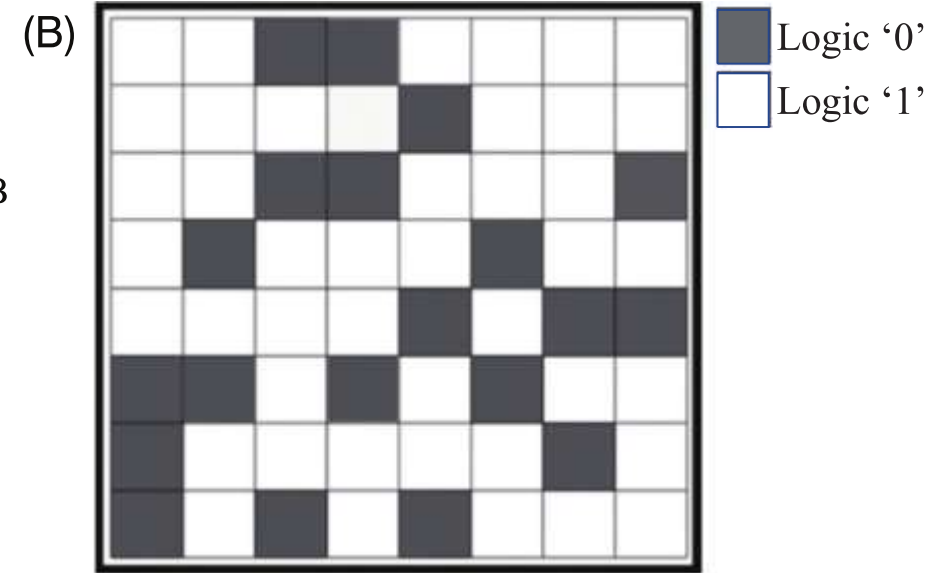
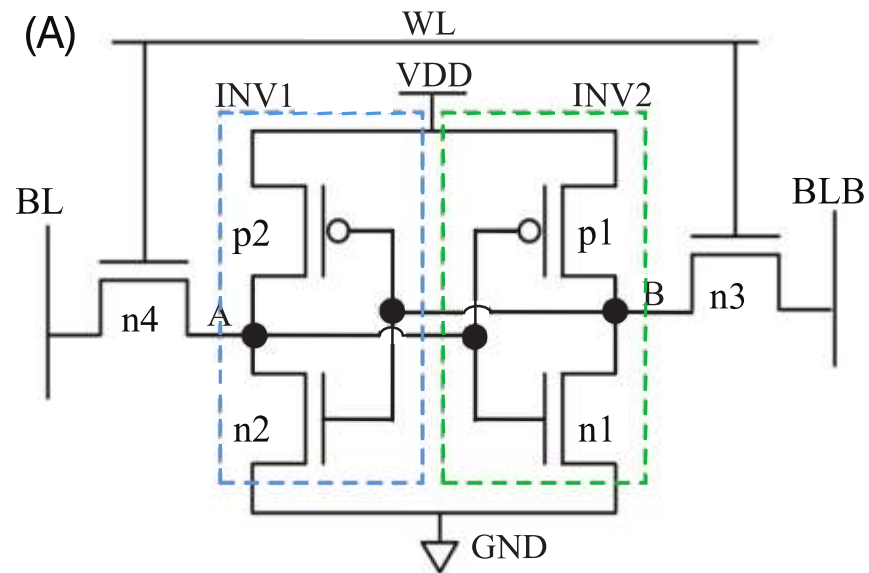
Lightweight PUF



VIA PUF



SRAM PUF



Emerging NVM Properties for PUF

Device	Manufacturing Variability Sources	Exploitable Features	Reliability Factors
PCM	- Geometric Variation (GST layer thickness and Bottom Electrode Contact Diameter) - Cell Resistance (R_{Cell}) - Write/Read Strength (required min. reset current) - Write current magnitude (I_{Write})	- Cell Variability - Stochastic Resistance Variation per cell - Programming sensitivity - variation in resistance level due to the nature of applied pulses	- Power supply Noise and Temperature Variation - Resistance Drift - Poor Endurance and Retention
Memristor & RRAM	- Variability in dope and undoped region length - Device thickness and Cross-sectional area - Stochasticity in doped and undoped resistance.	- Stochastic switching mechanism and intrinsic variability of devices. - Resistance variability due to applied voltage pulse duration	- Power supply Noise and Temperature Variation - Moderate Endurance and Retention - Read Disturbance
MRAM & STTRAM	- Geometric variation in free layer volume - Spin-torque Switching - Threshold voltage - Thermal Stability - Critical Switching Current	- Read Current variation - Meta-stability of free layer magnetization. - Variation in thermal energy - Back-hopping	- Temperature - External EM field - High Endurance and Retention